Graphs 2 — Equation of a line · Sketching · Simultaneous equations graphically

Squared paper needed for questions that say *draw*; a *sketch* is a rough drawing showing the intercepts and slope. Marks in brackets. ★ = harder question.

A · FINDING THE EQUATION OF A LINE

1 Find the equation of the straight line through $A(-2,\ -7)$ and $B(2,\ 5)$. [3]

2 A straight line passes through the origin O and the point $A(4,\ -2)$. [4]

a Find the equation of OA.

b Find the equation of the line parallel to OA that passes through $(-3,\ 1)$.

B · SKETCHING

3 Sketch these graphs, marking where each line crosses the axes. [6]

a $y = 2x - 3$ **b** $3x + 4y = 24$ **c** $y = -x + 1$

C · SIMULTANEOUS EQUATIONS GRAPHICALLY

4 Solve the simultaneous equations $x + y = 8$ and $y = 2x - 1$ graphically. Draw both axes from -2 to 8. [6]

5 At a stationery shop, 3 pens and 2 pencils cost €8, while 1 pen and 4 pencils cost €6. Let a pen cost x euros and a pencil y euros. Write a pair of simultaneous equations and solve them graphically to find the cost of one pen and of one pencil. [6]

Total: 25 marks

Check: c comes from substituting a point the line passes through · parallel means equal gradients · the solution of simultaneous equations is the crossing point — substitute it back into *both* equations.